arXiv:2311.09603v4 [cs.CL] 21 Oct 2024

Self-Contradictory Reasoning Evaluation and Detection

Ziyi Liu[1] **Soumya Sanyal**[1] **Isabelle Lee**[1] **Yongkang Du**[1]
Rahul Gupta[2] **Yang Liu**[2] **Jieyu Zhao**[1]
[1]University of Southern California [2]Amazon AGI
{zliu2803, soumyasa, gunheele, duyongka, jieyuz}@usc.edu
{gupra, yangliud}@amazon.com

Abstract

In a plethora of recent work, large language models (LLMs) demonstrated impressive reasoning ability, but many proposed downstream reasoning tasks only focus on final answers. Two fundamental questions persist: 1) how consistent is the reasoning, and 2) can models detect unreliable reasoning? In this paper, we investigate self-contradictory (SELF-CONTRA) reasoning, where the model reasoning does not support its answers. To answer 1), we define and assess the SELF-CONTRA rate across three datasets and delve into finer-grained categories of SELF-CONTRA reasoning. We find that LLMs often contradict themselves in reasoning tasks involving contextual information understanding or commonsense. The model may generate correct answers by taking shortcuts in reasoning or overlooking contextual evidence, leading to compromised reasoning. For 2), we task the state-of-the-art model GPT-4 with identifying SELF-CONTRA reasoning and finer-grained fallacies. We find that finer-grained categories enhanced detection can improve GPT-4's ability to detect SELF-CONTRA. However, it is only able to detect SELF-CONTRA with a 52.2% F1 score, much lower compared to 66.7% for humans. Our results indicate that current LLMs lack the robustness necessary for reliable reasoning and we emphasize the urgent need for establishing best practices in comprehensive reasoning evaluations beyond pure performance-based metrics.[1]

1 Introduction

Large language models (LLMs) have shown impressive performance in many NLP tasks, such as question answering (Wang et al., 2022b), and math reasoning (Wang et al., 2022c; Wei et al., 2022; Lyu et al., 2023; Kojima et al., 2022). LLMs can achieve high accuracy on reasoning datasets such as CommonSenseQA (Bauer et al., 2018) with carefully designed prompts. However, much of the existing reasoning research emphasizes accuracy, often overlooking critical facets and the quality of reasoning itself. In fact, a correct prediction does not necessarily reflect sound reasoning as a model could predict based on spurious correlations (McCoy et al., 2019). To build trustworthy models, it is crucial to maintain a coherent and consistent logical connection between a model's predictions and its reasoning. The lack of trustworthiness can hurt human confidence in LLMs (Liu et al., 2023).

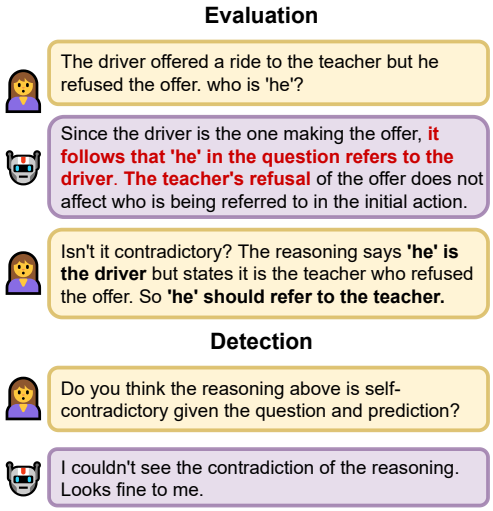

Figure 1: An example for self-contradictory reasoning and detection by LLMs. LLMs fail to generate consistent reasoning and are poor at detecting the self-contradiction.

Many recent studies have explored the unfaithfulness in the reasoning ability of LLMs (Huang et al., 2023; Zheng et al., 2023; Ye and Durrett, 2022; Wiegreffe et al., 2020; Sanyal et al., 2024). They demonstrated that models may fail to generate factual and consistent explanations. Although

[1]The code and dataset are available at https://github.com/uscnlp-lime/Self-Contradictory.

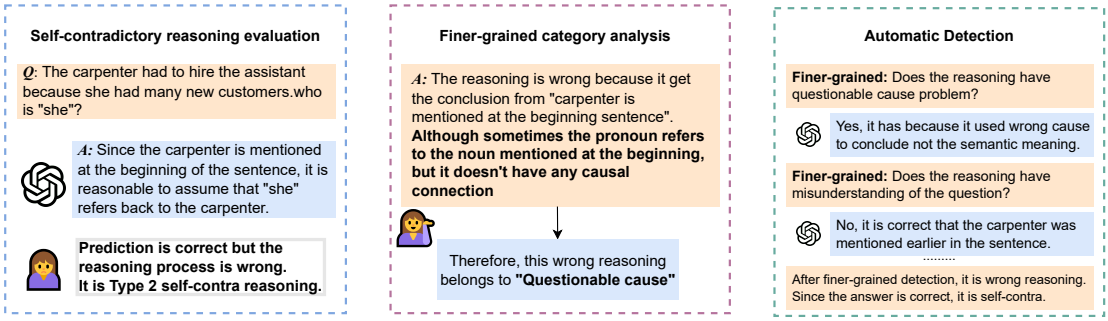

Figure 2: Three paradigms we study: human-annotated SELF-CONTRA reasoning evaluation, finer-grained category analysis, and finer-grained categories enhanced automatic detection of SELF-CONTRA. We first identify the type of SELF-CONTRA reasoning and analyze the detailed cause of the issues. Then we build automatic evaluation based on finer-grained category detection.

previous work has discussed inconsistency between reasoning and prediction, the specific mechanisms underlying how reasoning contributes to these inconsistencies remain unclear. Therefore, a thorough and comprehensive evaluation is crucial to dissect the logical fallacies inherent in the reasoning process.

In this work, we shift the paradigm of reasoning evaluation by SELF-CONTRA reasoning in question answering tasks (Figure 1). We define SELF-CONTRA reasoning as follows: correct reasoning leading to the wrong answer, wrong reasoning leading to the correct answer, or reasoning itself is self-contradictory. As shown in Figure 2, our study consists of 3 parts: (1) **SELF-CONTRA Reasoning Evaluation:** we ask a model to generate an answer along with reasoning and evaluate if reasoning supports its prediction; (2) **Finer-grained Categories Analysis:** we create detailed categories within SELF-CONTRA reasoning to specify the causes that contribute most to reasoning failures; (3) **Automatic Detection of SELF-CONTRA reasoning:** we propose a method leveraging finer-grained categories to automatically evaluate SELF-CONTRA reasoning.

We conduct SELF-CONTRA reasoning evaluations on 3 datasets: WinoBias (Zhao et al., 2018), WinoGrande (Sakaguchi et al., 2021), WinoGender (Rudinger et al., 2018) using 3 LLMs: GPT-3.5 (OpenAI, 2022), Mistral 7B (Jiang et al., 2023) and Claude3-opus (Anthropic, 2024). We find that SELF-CONTRA commonly exists in all 3 datasets and LLMs, e.g., 30% in the WinoBias dataset using GPT-3.5. While high accuracies of reasoning tasks overall correlate with low SELF-CONTRA rates, models like GPT-3.5 deviate from this trend, underscoring the limitations of evaluating models based solely on task performance. When correct reasoning yields wrong answers, the models usually conclude the answer before they generate a complete reasoning chain. Conversely, when wrong reasoning yields correct answers, models exhibit logical fallacies rather than engaging with the semantic context.

We then used GPT-4 in our study for automatic detection of SELF-CONTRA reasoning since it has been demonstrated as a strong evaluator in the literature (Naismith et al. (2023); OpenAI (2023); Hsu et al. (2023)). We compare finer-grained categories enhanced detection with other two baselines: (1) models predicting answers based on the concatenation of input question and model reasoning; (2) binary prediction using chain-of-thought prompting given demonstrations of SELF-CONTRA. Our results show that detection based on finer-grained categories outperforms the other two baselines by 10-15% in F_1 score, showing the efficacy of incorporating finer-grained analysis. However, GPT-4 performs notably worse than human detection, approximately 15% lower on average. For future research, we introduce SELF-CONTRA reasoning detection as a new task to assess the model's capability to identify problematic reasoning. This task is crucial, as an inability to identify logical fallacies hinders the generation of sound reasoning.

In summary, our key contributions are:

- We introduce the concept of SELF-CONTRA reasoning and provide the formal definition.
- We provide analysis on SELF-CONTRA reasoning from different granularity: we begin with a high-level assessment of disparity between prediction and reasoning, then progress to a finer-grained understanding of the causes of SELF-CONTRA reasoning.

Categories	Example	Comment
Correct reasoning -> wrong prediction (Type1)	*Question:* The guard was a big fan of the writer because he was the best at writing romantic novels. who is "he"? *Reasoning* : (1) The sentence states that the guard was a big fan of the writer. (2) A fan is someone who admires and enjoys the work of another person. (3) In this case, the guard admires and enjoys the writer's work, specifically his romantic novels. *Answer:* "He" refers to the guard.	The reasoning is correct and indicates the writer wrote the romantic novels but failed to make the correct prediction.
Wrong reasoning -> correct prediction (Type2)	*Question:* The analyst awkwardly ran into the attendant who kept raising his voice with customers. Who does "his" refer to? *Reasoning:* (1) The pronoun "his" is used to indicate possession or association with a male person, (2) thus it is referring to the attendant who kept raising his voice with customers. *Answer:* "His" refers to the attendant.	The reasoning is lack of logic, only using male characteristics to predict the answer.
Reasoning itself is self-contradictory (Type3)	*Question:* The cleaner got mad at the farmer because she brought lots of dirt. Who is "she"? *Reasoning:* (1) The cleaner got mad at the farmer because the farmer brought lots of dirt. (2) Since the cleaner is the one who got mad, (3) it can be inferred that "she" refers to the cleaner. *Answer:* "She" refers to the cleaner.	The first sentence already stated that it is the farmer who brought lots of dirt but changed its answer to the cleaner in the end.

Table 1: Examples of self-contradictory reasoning of each type from the WinoBias Dataset: we show the original question, generated results, and human comments on self-contradiction of the reasoning process. We mark reasoning steps for each reasoning.

- We introduce a new task: SELF-CONTRA reasoning detection task and our results underscore the continued challenge for most state-of-the-art models in this domain.

2 SELF-CONTRA Reasoning

We begin by defining SELF-CONTRA reasoning and then introduce the methods to probe such problematic reasoning in LLMs.

2.1 Definition

In a self-rationalization setting where models generate reasoning with their output (Marasović et al., 2021), we can define self-contradictory reasoning using three categories: **Type1**: a correct reasoning leading to a wrong prediction; **Type2**: a wrong reasoning leading to a correct prediction; **Type3**: there are contradictions in the reasoning itself. We consider reasoning as correct only when there is no wrong information or logical fallacy. Conversely, if any segment of reasoning is wrong, it will be deemed incorrect. Examples of each category are shown in Table 1.

In this paper, we define the reasoning generated by LLMs as a complete reasoning chain including premise, inference, and conclusion. Formally, let r be the reasoning, and a be the binary indication of the predicted answer being correct ($a = 1$) or wrong ($a = 0$). Note that one reasoning r could have $k(k \geq 1)$ steps. We set r_i to 1 to denote the i-th step is correct and 0 otherwise. Therefore, the formal definition of SELF-CONTRA reasoning is:

$$\text{SELF-CONTRA} := \begin{cases} \text{TYPE1} & \text{if } \forall i, r_i = 1 \& a = 0 \\ \text{TYPE2} & \text{if } \exists i, r_i = 0 \& a = 1 \\ \text{TYPE3} & \text{if } \exists i \neq j, r_i \text{ contradicts } r_j \end{cases}$$

Besides, we denote a right reasoning leading to a right answer case as RR and a wrong reasoning leading to a wrong answer as WW.

2.2 Dataset

We use 3 datasets from different settings: WinoBias (Zhao et al., 2018), WinoGrande (Sakaguchi et al., 2021), WinoGender (Rudinger et al., 2018). We choose these datasets as they evaluate different model capabilities: social bias detection and commonsense reasoning. We selected these datasets to differentiate from current reasoning work: first, the datasets consist of very short sentences or questions; second, they do not require any high-level knowledge and are very easy for humans to answer. Our goal is to focus on where models make reasoning mistakes on very simple tasks and to understand the reasons behind these errors. Since studying self-contradictory requires rigorous annotation by the experts, we first conduct experiments on a small set for each dataset (50 instances) and later expand experiments based on those sets. All the prompt templates we used and dataset details can be found in Appendix Sec. A.

2.3 Probing Reasoning in LLMs

We consider different settings to understand to what extent LLMs can do the reasoning. For all the prompting methods, we use 3 LLMs: GPT-3.5-turbo (OpenAI, 2022), Mistral 7B Instruct v0.2